THE ORDER OF ELIMINATING BLINDS IN MAZE LEARNING BY THE RAT[1]

KENNETH W. SPENCE

Laboratories of Comparative Psychobiology, Yale University

One of the most intriguing problems set by the experiments on animal maze learning is that of explaining or accounting for the order of difficulty of the blind alleys in a maze. The following paper is an attempt to throw some light upon the nature of the factors that are operating in the maze situation which determine this order. Before proceeding to this task, however, a brief review of the past studies concerned with this problem will be given. The primary purpose of this review is to summarize the outstanding experimental facts that have been obtained, and to discuss certain of the interpretations already given them.

I

The early studies of maze learning, from the point of view of the present problem at least, were complicated by the fact that the mazes used were not uniform in pattern. That is, the various culs-de-sac differed considerably from one another both as to their nature and as to the manner in which they digressed from the true path. The influences of these intra-pattern variations are revealed in such studies as those of Hubbert (7), Hubbert and Lashley (8), Peterson (13), White and Tolman (19), Dashiell (3), and Warden (16). Warden (16), for example, in an analysis of the Carr maze showed that there were three distinct types of culs-de-sac which differed considerably in difficulty. In spite of these complicating factors there appeared in several of these studies a tendency for fewer errors to be made at the food-box end of the

[1] This article developed out of a paper presented in the Seminar of Professor Clark L. Hull. The writer is indebted to Professor Hull for his critical reading of the manuscript.

maze, which led to the conclusion that the maze is learned from the food-box backward, there being a regressive order of elimination of errors. Carr (2) and Peterson (13), in particular, argued for this point of view and cited evidence in support of it.

With the development of more homogeneous maze patterns such as the Warden U pattern, the Stone multiple T pattern and more recently the Warner-Warden standard unit animal maze (18), many of the variable factors resulting from differences in culs-de-sac within the maze have been eliminated. The ruling out of these irrelevant variables has simplified the problem of interpretation to a considerable extent.

Outstanding in the recent experimental work in which these more uniform maze patterns have been employed is the establishment of the fact that blind alleys pointing in the direction of the goal (food-box) are entered more frequently than blinds pointing away from the goal and are also harder to eliminate. The notion that an orientation towards the goal plays an important part in maze learning is not altogether a recent one as is evidenced by the fact that Hubbert and Lashley (8), in 1917, suggested that the elimination in the Watson circular maze of those errors which consist in running past the right opening was probably due to a general orientation on the part of the animal toward the center where the goal was located. Recent investigations have served to emphasize to a still greater extent the importance of this direction orientation. Dashiell, in one of the most important studies involving this factor has shown that rats, on running a maze, soon develop this orientation in the general direction of the goal. He presents evidence showing that it plays a significant rôle in the learning of the maze. Studies by Yoshioka (20), Gengerelli (6), Tolman and Honzik (15a, 15b) have confirmed Dashiell's findings, all showing the operation of such a goal-orientation tendency.

The evidence of recent studies upon the question as to whether the proximity of the culs-de-sac to the goal end is a factor determining their ease of elimination tends to substantiate the earlier results of Peterson and Carr. There certainly is a general tendency, which is decidedly strong in the case of some mazes, for the

blind alleys farthest from the goal to be eliminated with greater difficulty than those nearest to the goal. The evidence for this conclusion is shown in table 1, which gives the results of several of the most recent investigations with homogeneous maze patterns. The values in the table were obtained as follows: In each study the blinds of the maze were ranked in order of difficulty, the easiest being ranked 1, the next easiest 2 and so on. Each maze was then divided into three parts and the average of the ranks of the blinds in each third of the maze computed. The

TABLE 1

Relative difficulty of sections of maze

INVESTIGATOR	TYPE OF MAZE	NUMBER OF BLINDS	SECTION OF MAZE		
			1st	2nd	3rd
Tolman (15a)	Multiple T	14	10.8	6.7	4.8
Tolman (15b)	Multiple T	14	10.8	6.7	4.8
Ruch (14)	Multiple T	12	10.2	4.7	4.5
Ruch (14)	Multiple T	12	9.2	4.7	5.4
Warden (17)	Warner-Warden	10	8.3	4.7	3.7
Husband (12)	Warden U	10	7.0	6.4	2.8
Warden (17)	Warden U	8	6.3	3.5	3.3
Warden (17)	Warner-Warden	8	6.3	3.0	3.7
Corey (2)		8	7.0	4.5	2.0
Warden (17)	Warner-Warden	6	5.5	2.0	3.0
Dashiell (5)	Warden U	6	5.5	3.0	2.0
Dashiell (5)	Warden U	6	5.0	2.5	3.0
Mean			7.66	4.36	3.58
S.D. Mean			.588	.447	.306

values in the table are these rank averages. It will be seen that the more difficult blinds (high values) tend to be in the first part of the maze and the less difficult blinds in the latter part, that is, near the goal. At the foot of the table the mean and the standard error of the mean of each section of the mazes are given. Table 2 shows the reliability of the differences between these means. The critical ratios $\left(\frac{\text{Diff.}}{\text{S.D. diff.}}\right)$ indicate that the chances are 100, 100 and 97, respectively, in 100 that the differences are true ones.

In a report of a study with the Warner-Warden standard unit maze, Warden and Cummings (17) have denied that there is any regressive order principle operating. Their own results, however, offer fairly good evidence that there is such a factor present as may be seen from table 3, which is a portion of Warden's table 4. The two shortest mazes, it is to be noticed, give a perfect regressive order and the three longer ones show a strong tendency towards such elimination. Yet in spite of this evidence

TABLE 2

Showing the reliability of the differences between the average rankings of the three sections

SECTIONS	DIFFERENCE BETWEEN THE MEANS	STANDARD DEVIATION OF THE DIFFERENCE*	DIFFERENCE / S.D. DIFF.
1st and 3rd	4.08	.389	10.49
1st and 2nd	3.30	.346	9.54
2nd and 3rd	.78	.403	1.93

* S.D. difference $= \sqrt{\sigma^2_{M_1} + \sigma^2_{M_2} - 2r\sigma_{M_1}\sigma_{M_2}}$.

TABLE 3

Order of elimination of culs-de-sac in the Warner-Warden maze

PATTERN	CULS-DE-SAC IN ORDER FROM ENTRANCE TO GOAL									
	I	II	III	IV	V	VI	VII	VIII	IX	X
2 culs	2	1								
4 culs	4	3	2	1						
6 culs	6	5	3	1	4	2				
8 culs	8	7	4	5	1	6	3	2		
10 culs	10	9	6	3	5	4	7	2	8	1

Warden writes (17, p. 252): "One theory seems to have been definitely exploded by these results. The notion that the animal maze is learned from the food-box backward (regressive order), which has long been held in some quarters without the slightest positive evidence of a crucial nature to support it, is now seen to be quite erroneous." His argument for this conclusion is that retracings, which were found to occur much more frequently in the section of the maze nearest the entrance compartment, thus

increasing the difficulty of that section, disturb the normal order of fixation. The fact that we do not get a perfect regressive order even with this added advantage of the retracing factor is, he claims, evidence against any regressive principle. The perfect regressive order found in the two shorter mazes (2 and 4 culs-de-sac) is dismissed on the "logical consideration" that these very short mazes are limiting cases of serial learning and would not exhibit the true laws of serial learning anyway.

Undoubtedly, as Warden argues, it is true that retracing tends to increase the errors in the blinds nearest the beginning of the maze. But this does not preclude the possibility that there might also be other factors operating that would produce such a result. Again, just because there is not a perfect order of elimination

TABLE 4

Distribution of mean number of forward errors per rat

PATTERN	CULS-DE-SAC IN ORDER FROM ENTRANCE TO GOAL									
	I	II	III	IV	V	VI	VII	VIII	IX	X
2 culs	8.80	6.00								
4 culs	10.44	5.66	4.44	1.66						
6 culs	8.44	5.33	2.33	2.11	3.33	1.55				
8 culs	14.90	6.30	3.10	2.40	1.90	2.30	3.10	2.00		
10 culs	14.25	7.00	4.63	1.25	2.38	1.88	2.65	1.00	3.88	1.0

does not necessarily mean that there is not existent some regressive order principle, for, as is evident upon a little reflective consideration, such a principle could be operating without there resulting perfect reverse elimination. Thus if there were any other factor, or factors operating simultaneously with a regressive order factor the reverse order could be disturbed though not completely obliterated. Especially would this be so if this second factor should be in no way related to the order of the blinds in the maze.

That another such factor actually was present in the experiment with the Warner-Warden maze can easily be shown. Warden's own results, in fact, do not justify his belief that all other factors save order (retracing excepted) were properly con-

trolled in this maze pattern. An examination of table 4, which is a part of Warden's table 3 will give some indication of what this factor is. It is to be noted in particular that the odd blinds tend to be more difficult than the even blinds. Thus, in the ten cul-de-sac pattern, blinds III, V, VII, and IX are all more difficult than the even blinds IV, VI, VIII and X. This fact of the greater difficulty of the odd blinds is shown very clearly in figure 1, which presents in graphical form the data (errors) for the 10

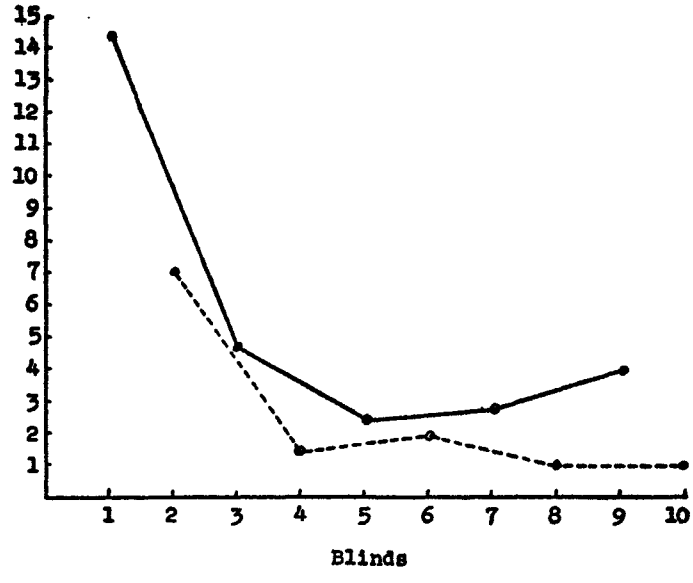


FIG. 1

culs-de-sac pattern. A solid line joins the odd blinds and a broken line the even blinds.

Now the maze pattern employed by Warden (see fig. 2) was a symmetrical linear one that involved a simple right-left alternation of turns. All the odd blinds involved making a *left* turn and all the even blinds a right turn. The final choice (to the goal) consisted in making a *left* turn, the same, it should be noticed, as that involved in the odd blinds which were the more difficult ones. We appear to have here a factor somewhat similar to the goal direction tendency referred to above. That it is probably

not the same thing, however, may be seen from the drawing of the maze. At each point of choice it is the true path that points in the direction of the goal. Neither group of blinds points definitely in the direction of the goal. This factor, then, might more aptly be described as an anticipatory goal reaction, considering the turning left into the food-box as a part of the total goal reaction. But whatever the nature of this factor may be, its presence and influence cannot be denied. Moreover, that it would tend to upset a perfect reverse order of elimination can readily be seen.

In summing up the evidence from the recent studies of maze learning, then, there are two outstanding facts: (*a*) Blind alleys pointing in the direction of the goal tend to be entered more often than those pointing away and are harder to eliminate; (*b*) there is a general tendency for the blinds to be more difficult the farther they are away from the goal.

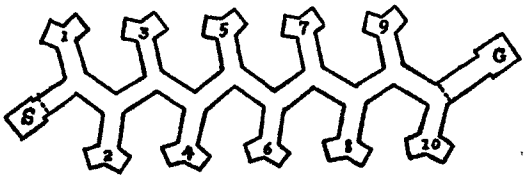


FIG. 2

II

From the preceding section it is evident that even in homogeneous maze patterns there is usually more than a single factor operating to determine the difficulty of the blinds. Any attempt, then, to account for the order of difficulty of the blind in maze learning by the operation of a single factor or principle, such as the primary-recency principle is not likely to prove very successful unless that factor plays an extremely important part compared with any others that are operating. The number of such factors and their importance will, in all probability, vary with the different maze patterns and even in the same maze under different experimental conditions.

The evidence from the experimental studies, as have been

indicated in the previous section, suggests that there are two particularly important factors operating in the maze situation determining the order of difficulty of the blinds; one of such a nature as to produce, if acting alone, a backward order of difficulty, and a second which operates to increase the difficulty of the blinds pointing in the direction of the goal. Hypothetical accounts of the nature of two such factors have been put forward in the past, as was intimated previously. Particularly is this so with respect to the former factor. Usually, however, they have been conceived in vague pleasure-pain terms with no satisfactory account of their mode of operation being given. Even when these attempts have been in more objective, physiological terms they have proved to be of little value as they did not lead to or even permit the setting up of critical experimentation.

Recently there has been put forward in connection with some theoretical considerations of rat learning an hypothesis from the assumptions of which it may actually be deduced that the blind alleys will be more difficult the farther they are away from the goal. This hypothesis, the goal gradient hypothesis of Hull (10), is, moreover, capable of explaining (deducing) a wide variety of rat behaviour phenomena that have already been experimentally established and also some that remain to be investigated, but which in general have never before been shown to possess any organic relationship to each other.

In the following theoretical attempt to account for the order of difficulty of the culs-de-sac in the maze the writer has assumed this hypothetical goal gradient mechanism as one of the factors determining this difficulty. A second factor to be assumed is some (hypothetical) mechanism that tends to produce a particularly strong tendency on the part of the animal to proceed, wherever possible, in a direction towards the goal; thus if at a given choice-point, one path leads directly towards the goal and one in the opposite direction the animal should, as the result of this mechanism tend to prefer the former path. Dashiell proposes a rather general hypothesis as to the possible mode of functioning of this factor. He suggests that, at the time the animal enters the maze a particular implicit process (set or posture) is

established or aroused, which persisting in part at least, operates as a guiding factor in the subsequent responses of the animal in the maze, tending to facilitate or strengthen any turns allied with it and to weaken any response antagonistic to it. Such a mechanism acting in a maze would tend to strengthen the excitatory tendencies to take the paths (right or wrong) leading in the absolute direction of the goal, and to weaken the excitatory tendencies to take the paths leading in the opposite direction. Any blind alley pointing in the absolute direction of the goal, then, would be eliminated with relatively considerable difficulty as the tendency to enter it is strengthened while the tendency to take the true path at this point is weakened. Especially is this so in the recent types of maze patterns, in which the true path and the blind alley point in opposite directions.

Hull (11), in an unpublished manuscript kindly made available to the author, has also developed a plausible account of the acquisition of this "Direction Orientation" with respect to the goal in his concept of the habit family hierarchy. It is sufficient to state here that this mechanism, like that suggested by Dashiell would result in a preference on the part of the animal for the path leading in the absolute direction of the goal over the one leading away. For our present purpose the description of this factor in terms of its assumed effects is sufficient.

Although the manner in which the hypothesis of the goal gradient is conceived as functioning need not concern us here in all its details some understanding of it is necessary. According to this hypothesis all stimuli, in particular the drive stimulus, get conditioned to every act of a sequence, the goal reaction being the most strongly conditioned and the other reactions progressively weaker as they are more remote (in time) from the goal reaction. The slope of this gradient Hull assumes to be positively accelerated, being more steep at the goal and becoming less and less steep as the goal becomes more and more remote. As he has shown, a logarithmic curve very accurately fits the experimental results of Yoshioka's careful experiment on discrimination of maze patterns (21). In the present discussion, however, we shall employ a gradient curve of the type

$$Y_N = Y_G(1-1/n)^N$$

in which Y_G stands for the strength of the excitatory tendency of the goal reaction, $1/n$ for the fraction that this tendency becomes weakened at each unit distance from the goal and Y_N for the strength of the tendency N units from the goal.[2] Table 5 gives the strength of the excitatory tendencies at various unit distances from the goal assuming that Y has a strength of 10 points and that the excitatory tendency is weakened one-tenth of its value at each unit distance from the goal.

The manner in which Hull conceives this goal gradient hypothesis as mediating a backward order of elimination of blinds is as follows (10): Once food has been found and consumed the drive

TABLE 5

Strength of excitatory tendencies. $Y_N = 10(1-1/10)^N$

N	Y_N	N	Y_N
0	10.00		
1	9.00	11	3.13
2	8.10	12	2.82
3	7.29	13	2.54
4	6.56	14	2.285
5	5.90	15	2.056
6	5.31	16	1.851
7	4.78	17	1.676
8	4.30	18	1.508
9	3.87	19	1.358
10	3.48	20	1.222

stimulus becomes conditioned to the goal reaction and the several phases of the motor sequence leading to the goal reaction with an intensity progressively weaker according as each is farther away (in time) from the goal. Thus in the maze in figure 3, in which each blind and each segment of the true path is considered as being one unit in length, the excitatory tendency to each reaction will be as represented by the figure in the bracket. That is, at the point F the turning movement to the left (true path) is one unit from the goal reaction. From table 5 it will be seen that

[2] This curve is positively accelerated, the slope becoming greater as the goal is approached.

the value of an excitatory tendency of a reaction one unit from the goal is 9.00 points. At the same choice-point, *F*, a turning movement to the right, requiring as it does going into and coming out of the blind alley, involves two units more of time and effort. That is, this right turning movement is about two units farther from the goal reaction than the action of turning to the left, making it three units in all from the goal reaction. This gives its excitatory tendency a strength of 7.29 points (table 5), making a difference of 1.71 points in favor of the left turn, the one into

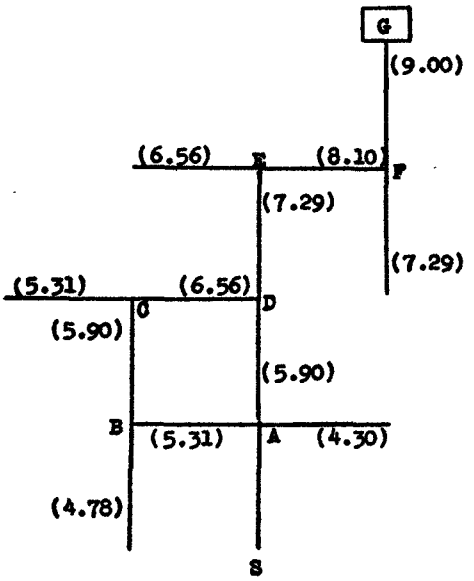


FIG. 3

the true path. This difference in favor of the true path results ultimately in the elimination of the blind alley.

It is to be noted further that the differences between the true path and the blind alley reactions at each point of choice become less and less the farther they are away from the goal end of the maze. That is, the gradient difference available to mediate the elimination of blind alleys that are some distance from the goal is less than is available for blind alley elimination near the goal. This progressive weakening of the differential value available for

blind alley elimination as distance from the goal increases would result in a corresponding progressive difficulty in the elimination of blinds. That is, if this factor alone were operating to determine the order of difficulty of the blinds there would result a reverse order.[3]

III

Assuming the simultaneous operation of these two hypothetical mechanisms, the goal gradient and the direction orientation mechanism, let us attempt to deduce the order of difficulty that should occur on some particular maze on which there is available experimental data. The adequacy of these hypotheses to account for the experimental facts can readily be tested, then, by comparing the theoretically determined order with the actual experimental findings. The maze study we have selected for this purpose is one carried out by Tolman and Honzik (15a, 15b) concerned with reward and maze learning. Their study is particularly favorable for our purpose as they did not allow retracing in the maze and the pattern they used had no T's on the ends of the blinds. These complicating factors, then, are ruled out. A diagram of their maze is given in figure 4.

We shall begin by considering the goal gradient factor. Table 6 shows how the differential value in favor of the true path at each choice-point is derived. Column 1 lists the choice-points from 14 at the goal end to 1 at the start of the maze. The second column gives the unit distance of reactions into the true path and blind at each choice-point from the goal. The blind, it should be noted, always involves traversing 2 more units than the corresponding true path. The third column shows the rela-

[3] It should be noted that according to the goal gradient hypothesis the differences between the difficulty of the blind alleys at the goal end of the maze should be greater than these differences at the beginning of the maze. This implication of the hypothesis does not fit with the facts of table 1, in which there is shown to be a greater difference between the beginning and the middle section than between the goal and the middle section. These facts can be accounted for in some of the mazes as being due to the unequal distribution of goal-pointing alleys in the three sections; but even when this factor is held constant there is evidence that the differences do not follow exactly as demanded by the hypothesis. A difficulty is thus presented that the future development of the theory will have to take account of.

tive strength of the competing tendencies at each point of choice (based on table 5), and column 4 the difference between these two tendencies. This differential available for the elimination of the blind alley, it is to be noted, becomes less and less the farther away from the goal a choice is.

Now let us consider the direction orientation factor. This mechanism, as has already been explained, is conceived as tending to favor the reaction pointing in the absolute direction of the

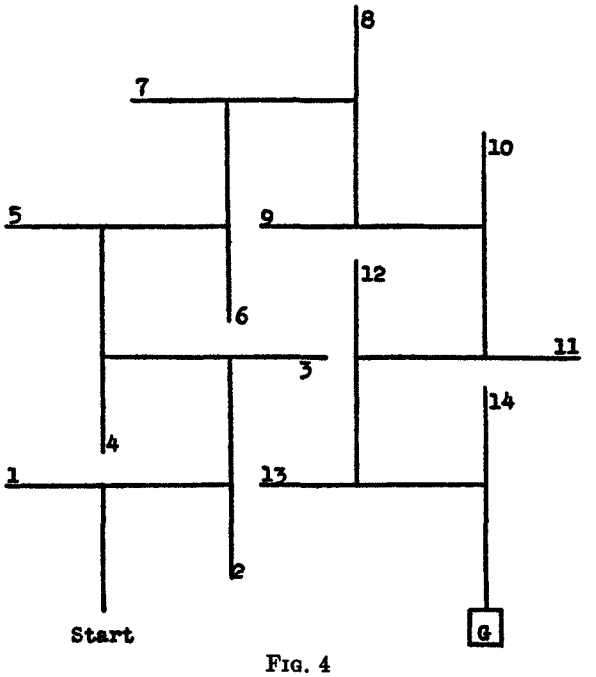


FIG. 4

goal over the reaction pointing away from the goal. Table 7 shows the direction of the true path and the blind path at each choice-point with respect to the goal, T standing for toward the goal and A for away.[4] The third column shows the effect of this orientation process upon the tendency to select the true path. We have assumed that at each choice-point in the maze there

[4] In this pattern all the blinds pointing toward the right and down point towards the goal.

TABLE 6

The derivation of the differential value in favor of true path based on goal gradient hypothesis

1	2		3		4
CHOICE POINT	UNIT DISTANCE FROM GOAL		RELATIVE STRENGTH OF COMPETING TENDENCIES		DIFFERENTIAL VALUE IN FAVOR OF TRUE PATH OVER BLIND
	T. path	Blind	T. path	Blind	
14	1	3	9.00	7.29	1.71
13	2	4	8.10	6.56	1.54
12	3	5	7.29	5.90	1.39
11	4	6	6.56	5.31	1.25
10	5	7	5.90	4.78	1.12
9	6	8	5.31	4.30	1.01
8	7	9	4.78	3.87	.91
7	8	10	4.30	3.48	.82
6	9	11	3.87	3.13	.74
5	10	12	3.48	2.82	.66
4	11	13	3.13	2.54	.59
3	12	14	2.82	2.285	.535
2	13	15	2.54	2.056	.484
1	14	16	2.285	1.851	.434

TABLE 7

1	2		3	4
CHOICE POINT	DIRECTION WITH RESPECT TO GOAL		EFFECT DIRECTION TENDENCY ON REACTION INTO TRUE PATH	RELATIVE STRENGTH OF COMPETING TENDENCIES AS DETERMINED BY TWO HYPOTHETICAL FACTORS
	T. path	Blind		
14	T	A	1.00	2.71
13	T	A	1.00	2.54
12	T	A	1.00	2.39
11	*A*	T	−1.00	.25
10	T	A	1.00	2.12
9	T	A	1.00	2.01
8	T	A	1.00	1.91
7	T	A	1.00	1.82
6	*A*	T	−1.00	−.26
5	T	A	1.00	1.66
4	*A*	T	−1.00	−.41
3	*A*	T	−1.00	−.465
2	*A*	T	−1.00	−.516
1	T	A	1.00	1.434

results from the operation of this factor an increase in strength of 1.00 in favor of the reaction pointing towards the goal over the reaction pointing away. The values in the third column, then, are positive in those cases in which the true path points towards the goal and negative when the true path points away. The unit 1.00, selected as the preferential value resulting from this mechanism was chosen because it is about midway between the highest and the lowest of the differential values resulting from the operation of the goal gradient factor, thus tending to give each equal weight. The direction orientation mechanism, it should be noted, is conceived as having the same degree of effectiveness throughout the maze, i.e., it operates as strongly at the beginning of the maze as near the goal.

The values in the fourth column of table 7 show the algebraic summation of the strengths of these two jointly operating tendencies at each choice point, i.e., they represent the combined effect of the two hypothetical factors. At some of these points the two factors are in harmony with each other, both favoring the excitatory tendency to take the true path over the tendency to enter the blind. At others, those in which the blinds point in the direction of the goal, they are opposed to each other, the direction orientation mechanism favoring the tendency to enter the blind and the goal gradient mechanism favoring the reaction of taking the true path. We see that the greatest difference in favor of the true path is at choice 14, the one nearest the goal. The blind at this point, then, we should deduce to be the easiest of the blinds to learn to avoid.

While the absolute values of the figures in this column have not a great deal of significance, the negative values are of some interest. These negative values would imply that at some time in the learning process there would actually be set up a tendency to prefer the blind at these points over the true path. Such tendencies are quite commonly observed in the learning of a maze; rats often persist in running into certain blind alleys for a considerable number of trials. Some other mechanism, then, such as inhibition resulting from frustration must be invoked to explain the final elimination of these blinds.

The first row of table 8 shows the order of difficulty of the blinds in this maze which we should expect to find if two such mechanisms as those assumed were the sole determiners, the easiest blind being given the rank 1, the next easiest 2 and so on. The second and third row of this table give the experimentally determined orders of difficulty of the blinds for two different groups of rats (15a, 15b). It will be seen that the order deduced from the two assumed principles or factors corresponds very closely with the experimental results, the correlations (rank difference) in the 2 cases being .90 ±.03 and .91 ±.03. These two hypothetical mechanisms operating jointly would seem, then, to account extremely well for the experimental results on this particular maze.

There are, undoubtedly, other factors than these two operating in this maze which are influential in determining the order

TABLE 8

Showing the theoretical and experimental orders of difficulty of the blind alleys

	NUMBER OF BLIND													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Theoretical	9	14	13	12	8	11	7	6	5	4	10	3	2	1
Experimental	9	14	13	10	8	12	4	6	5	3	11	2	7	1
	9	14	13	10	8	12	4	5	6	2	11	3	7	1

of difficulty of the blinds. There is evidence in the experimental literature of such factors. Dashiell and Bayroff (5), for instance, have shown that turns made at choices encountered later in the maze are subject to the influence of preceding choices even in the first trial. Anticipatory tendencies have also been shown to be a factor in determining errors in the maze. In connection with this last type of factor there may be mentioned the hypothesis of the anticipatory goal reaction suggested by Hull (9). According to this hypothesis the goal reaction, or a fractional component of it, tends with practise in the situation to intrude into the very beginning of the behaviour sequence and to persist throughout the cycle. The operation of such a mechanism might be conceived as tending to increase the number of errors in those blinds involving the same direction turn, right or

left, as the last turn leading directly to the goal. That is, this last turn might be considered a fractional part of the goal reaction. On the assumption that such a mechanism was operating in Tolman's maze along with the two we have been considering, the theoretical order resulting from all three mechanisms operating jointly correlated slightly higher, .93, with the experimental results than the order deduced from only two.

As has been pointed out the conditions under which Tolman's experiment took place (no retracing and the non-complication of the blind alleys with T's) were particularly favorable to our pres-

TABLE 9

INVESTIGATOR	TYPE OF MAZE	NUMBER OF BLINDS	WEIGHT OF DIRECTION FACTOR	GOAL GRADIENT	GOAL GRADIENT AND DIRECTION FACTOR
Ruch (14)	Multiple T	12	1.00	.63	.77
Ruch (14)	Multiple T	12	1.00	.40	.36
Warden (17)	Warner-Warden	10	.10	.48	.77
Warden (17)	Warner-Warden	8	.10	.52	.50
Warden (17)	Warner-Warden	6	.10	.71	.77
Husband (12)	Warden U	10	.10	.61	.45
Dashiell (5)	Warden U	6	.10	.80	.80
Dashiell (5)	Warden U	6	*	.60	
Corey (2)	No diagram	8		.88	

* Blind alleys all pointed in the same direction.

ent purpose, that of determining the extent to which the two above assumed factors are able to account for the order of difficulty of the blind alleys. An examination of the literature failed to reveal any other studies with data bearing upon our problem in which these complicating factors were altogether eliminated. Table 9, however, presents the results of an analysis of what data are available as to the difficulty of blind alleys in the more recent homogeneous maze patterns. The first three columns of this table give information as to the investigator and the type of maze pattern used. The fourth column shows the weight given the direction factor in each maze. In the multiple T mazes the

value is 1.00, while in the Warden type mazes this value is only .10 as these patterns are uni-directional, i.e., the goal is neither to the right or to the left but straight ahead. In these uni-directional mazes the mechanism operating is assumed to be that of the anticipatory goal reaction. The fifth column gives the correlation (rank method) between the experimental order and the theoretical order as deduced from the operation of the goal gradient factor alone, and the sixth column the correlation between the experimental results and the theoretical order as deduced from the joint operation of the goal gradient and the direction orientation, or the anticipatory goal reaction in the case of the uni-directional mazes. The correlations, it will be seen, are all positive and with one or two exceptions fairly high. The possibility that such consistently high positive values are merely the result of chance is extremely remote.

SUMMARY

This paper has been concerned with the nature of the factors operating in the maze situation that are responsible for the order of difficulty of the blind alleys, and also, with ascertaining the extent to which certain hypotheses that have been put forward in connection with this problem are capable of accounting for the facts. The following conclusions have been reached:

1. There is considerable evidence of some backward order principle of learning operating in the maze situation. In general, the blinds are more difficult the farther they are from the goal.

2. Such a principle alone, however, is not able to account for the actual order of blind alley elimination.

3. Other factors that appear to be of importance in determining the order of difficulty of blind alleys are an absolute direction orientation on the part of the animal to the goal and a tendency to anticipate, particularly the goal reaction.

4. When the two or three major factors are known, there is reason to believe that the order of difficulty of the blinds in a maze may be deduced or predicted with considerable precision.

REFERENCES

(1) CARR, H.: The distribution and elimination of errors in the maze. Jour. Animal Behav., 1917, vii, 145–159.
(2) COREY, S. M.: An experimental study of retention in the white rat. Jour. Exper. Psychol., 1931, xiv, 252–259.
(3) DASHIELL, J. F.: The need for an analytic study of the maze problem. Psychobiol., 1920, ii, 181–186.
(4) DASHIELL, J. F.: Direction orientation in maze running by the white rat. Comp. Psychol. Monog., 1930, vii, no. 2.
(5) DASHIELL, J. F., AND BAYROFF, A. G.: A forward-going tendency in maze running. Jour. Comp. Psychol., 1931, xii, 77–95.
(6) GENGERELLI, J. A.: The principle of maxima and minima in animal learning. Jour. Comp. Psychol., 1931, xi, 193–236.
(7) HUBBERT, H. B.: Elimination of errors in the maze. Jour. Animal Behav., 1915, v, 66–72.
(8) HUBBERT, H. B., AND LASHLEY, K. S.: Retroactive association and the elimination of errors in the maze. Jour. Animal Behav., 1917, vii, 130–138.
(9) HULL, C. L.: Goal attraction and directing ideas conceived as habit phenomena. Psychol. Rev., 1931, xxxviii, 487–506.
(10) HULL, C. L.: The goal gradient hypothesis and maze learning. Psychol. Rev., 1932, xxxix, 25–44.
(11) HULL, C. L.: Habit family hierarchy. (In manuscript.)
(12) HUSBAND, R. W.: Comparison of human adults and white rats in maze learning. Jour. Comp. Psychol., 1929, ix, 9, 361–379.
(13) PETERSON, J.: The effect of length of blind alleys on maze learning: An experiment on twenty-four white rats. Behav. Monog., 1917, iii, no. 15, 53 p.
(14) RUCH, F. L.: Food-reward vs. escape-from-water as conditions motivating learning in the white rat. Jour. Genet. Psychol., 1931, xxxviii, 127–145.
(15a) TOLMAN, E. C., AND HONZIK, C. H.: Degrees of hunger, reward and nonreward, and maze learning in rats. Univ. of California Pub. in Psychol., 1930, iv, 241–256.
(15b) TOLMAN, E. C., AND HONZIK, C. H.: Introduction and removal of reward, and maze performance in rats. Univ. of California Pub. in Psychol., 1930, iv, 257–275.
(16) WARDEN, C. J.: Some factors determining the order of elimination of culs-de-sac in the maze. Jour. Exper. Psychol., 1923, vi, 192–210.
(17) WARDEN, C. J., AND CUMMINGS, S. B.: Primacy and recency factors in animal motor learning. Jour. Genet. Psychol., 1929, xxxvi, 240–257.
(18) WARNER, L. H., AND WARDEN, C. J.: The development of a standardized animal maze. Arch. Psychol., 1927, xv, no. 93.
(19) WHITE, A. E., AND TOLMAN, E. C.: A note on the elimination of short and long blind alleys. Jour. Comp. Psychol., 1923, iii, 327–331.
(20) YOSHIOKA, J. G.: Direction as a factor in maze solution in rats. Jour. Genet. Psychol., 1930, xxxviii, 307–320.
(21) YOSHIOKA, J. G.: Weber's law in the discrimination of maze distance by the white rat. Univ. of California Pub., 1929, iv, 155–184.